Bioenergetics

Complete study guide: every slide explained in plain English, with jot-worthy notes, worked examples and practice questions

HOW TO USE THIS GUIDE (so you stop copying whole slides)

1. Read the **Plain English** boxes first. That is the understanding.

2. Copy **only** the yellow **JOT THIS DOWN** boxes into your notebook. Together they are your revision sheet.

3. Do **Test Yourself** at the end without looking, then check the answers.

4. Red boxes are the mistakes lecturers love to set questions on.

What is inside

1	The big picture: bioenergetics, metabolism, ATP, energy charge
2	Free energy and the laws of thermodynamics
3	State functions G, H, S and the Gibbs equation (with worked examples)
4	Redox potential
5	Exergonic and endergonic reactions, coupling
6	High-energy compounds and their classification
7	ATP: structure, energy, functions
8	Corrections and traps found in the slides
9	Test yourself with full answers
10	One-page revision sheet

PART 1 | The Big Picture: Bioenergetics, Metabolism and ATP

1.1 What is bioenergetics?

THE IDEA IN PLAIN ENGLISH

Bio = life. **Energetics** = study of energy. So **bioenergetics = the study of energy changes in biological (living) reactions**.

It is the part of biochemistry that asks: **how do living cells obtain energy, change it from one form to another, and use it?** The slide gives the "goal" as describing how organisms **acquire, transform and utilize** energy to perform **biological work** (moving, building molecules, pumping ions, growing, reproducing).

1.2 Metabolism: catabolism and anabolism

THE IDEA IN PLAIN ENGLISH

Metabolism = **all the chemical reactions** happening in an organism. Studying metabolic pathways is essential to bioenergetics.

Think of the cell's energy like a bank account. **Catabolism is income** (breaking food down releases energy, which is banked as ATP). **Anabolism is spending** (building proteins, DNA and so on costs energy, paid in ATP). **ATP is the cash** that moves between the two.

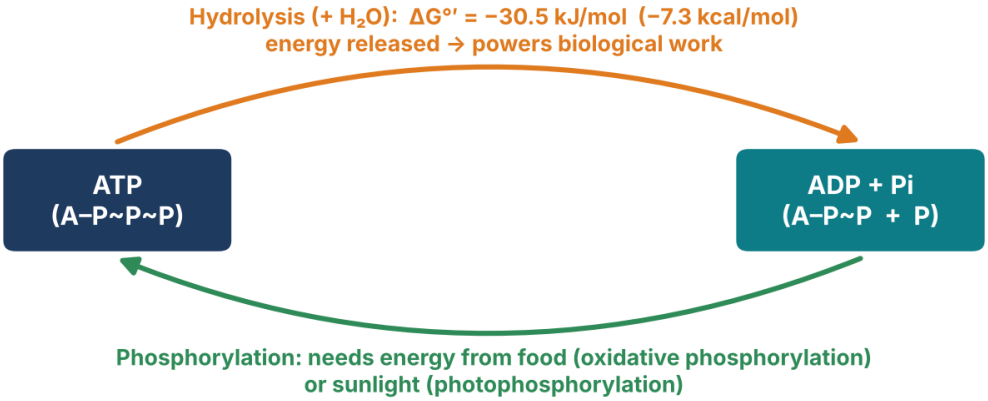
	Catabolic reactions	Anabolic reactions
What happens	**Breakdown** of chemical molecules (big → small)	**Synthesis** of compounds (small → big)
Energy	Releases energy (mostly exergonic)	Needs energy (mostly endergonic)
ATP	**Makes** ATP	**Uses** ATP
Examples	Cellular respiration, glycolysis, digestion	Photosynthesis, protein synthesis, DNA synthesis

1.3 ATP: the energy currency

THE IDEA IN PLAIN ENGLISH

Adenosine triphosphate (ATP) is the main "energy currency". You do not pay for things in cassava or yams; you use naira. In the same way, cells do not run reactions directly on glucose; they convert glucose energy into ATP, then spend ATP wherever energy is needed.

The slide gives two goals of metabolism: (1) **synthesize ATP** from starting materials taken from the environment, and (2) **break ATP down** (into **ADP** + inorganic phosphate, Pi) by using it to drive biological processes.



The ATP–ADP cycle. Breaking ATP down releases usable energy; rebuilding it costs energy.

1.4 Energy charge

THE IDEA IN PLAIN ENGLISH

The slide defines **energy charge** as the **ratio of ATP to ADP** in the cell. It acts like a fuel gauge that tells the cell what to do:

- **More ATP than ADP** (cell is "full") → the cell **can use ATP to do work**.
- **More ADP than ATP** (cell is "low") → the cell **must make ATP** through **oxidative phosphorylation** (the ATP-making process in respiration).

BEYOND THE SLIDE (still examinable)

Formal definition (Atkinson): energy charge = ([ATP] + ½[ADP]) / ([ATP] + [ADP] + [AMP]). It ranges from 0 to 1, and healthy cells keep it around 0.8 to 0.95. The slide's "ATP/ADP ratio" is the simplified version, so use the slide's wording but know this exists.

Three ways cells make ATP: substrate-level phosphorylation (a high-energy compound hands its phosphate to ADP, e.g. glycolysis), **oxidative phosphorylation** (using the energy from electron transport, in mitochondria), **photophosphorylation** (using light energy, in chloroplasts).

JOT THIS DOWN

Bioenergetics = study of energy changes/transformation and use in living cells. Goal: how organisms **acquire, transform, utilize** energy for biological work.

Metabolism = all chemical reactions in an organism. **Catabolic** = breakdown (releases energy, makes ATP). **Anabolic** = synthesis (uses energy, uses ATP).

ATP = main energy currency. Goals: make ATP; break ATP → ADP + Pi to do work.

Energy charge = ATP : ADP ratio. More ATP → use it. More ADP → make ATP via **oxidative phosphorylation**.

PART 2 | Free Energy and the Laws of Thermodynamics

2.1 Free energy

THE IDEA IN PLAIN ENGLISH

Staying alive is hard work: cells must do work to stay alive, grow and reproduce. Where does the energy come from? Not all the energy in a molecule can be used. The part that **can actually be used to do work** is called **free energy** (the slide: "the energy actually available to do work (utilizable)").

Think of your salary: the gross amount is bigger than what you can really spend after deductions. Free energy is the "take-home pay" of a reaction.

- **ΔG** (change in free energy) is valuable for **predicting whether a reaction is feasible**.
- **A reaction can occur spontaneously if it is accompanied by a decrease in free energy** (ΔG negative). Reactions roll "downhill" in free energy, like water flowing downhill.
- During a chemical reaction, **heat may be released or absorbed** (this leads to enthalpy, Part 3).

WATCH OUT: EXAM TRAP

"Spontaneous" does NOT mean "fast". It only means the reaction is thermodynamically allowed (no net energy input needed). A spontaneous reaction can be very slow. Enzymes make reactions faster, but **they never change ΔG**.

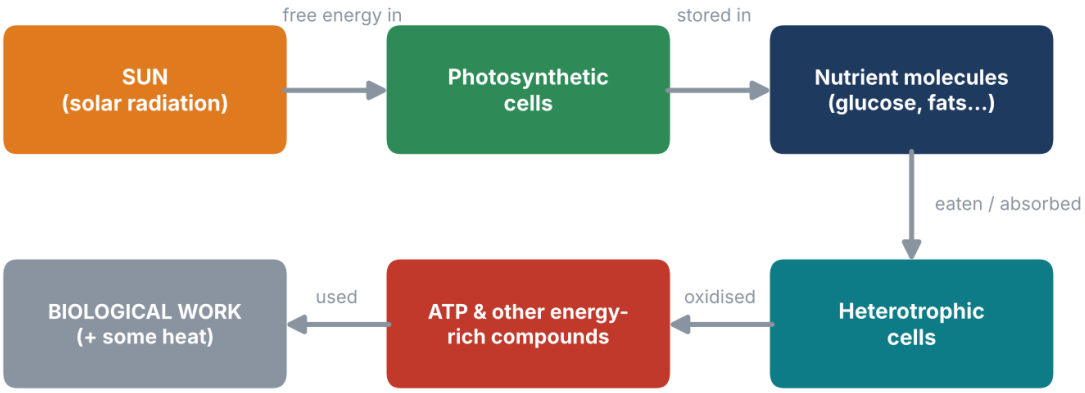
2.2 How cells get free energy

THE IDEA IN PLAIN ENGLISH

Cells are isothermal systems: they work at a **constant temperature and pressure**. This matters because heat can only do work when it flows from hot to cold, and a cell has no big temperature difference inside it. So cells **cannot run on heat**; they must use **chemical free energy**.

Cell type	Where it gets free energy	Examples
Photosynthetic cells	Absorbed **solar radiation** (light)	Plants, algae, cyanobacteria
Heterotrophic cells	**Nutrient molecules** (food)	Animals including humans, fungi, most bacteria

Both then **transform this free energy into ATP and other energy-rich compounds** to provide energy for biological work.



Energy flow in the living world: sunlight → nutrients → ATP → work (some energy always leaves as heat).

2.3 The two laws of thermodynamics

Law	Plain meaning	In the cell
1st law = conservation of energy	Energy is never created or destroyed. It only changes form or moves from one place to another. Total energy of a closed system stays constant.	Chemical energy in glucose → ATP → mechanical (muscle), heat, etc. Nothing is lost, it only changes form.
2nd law = the universe tends toward increasing disorder	In every natural process the **entropy** (disorder) of the universe increases.	Every reaction releases some energy as heat, which spreads out and increases disorder. This is why you feel warm after exercise.

THE IDEA IN PLAIN ENGLISH

A tidy room naturally becomes messy, but a messy room never tidies itself. That is the 2nd law. Cells are very orderly, so how do they get away with it? They **spend energy (ATP) to build order inside themselves** and in doing so release heat and waste that increase disorder outside. The **total** entropy of cell + surroundings still goes up. That is what the slide's "unless energy-requiring processes counteract it" means: a *local* decrease in disorder is possible, but only by paying with energy.

BEYOND THE SLIDE (still examinable)

System vs surroundings: the system is the part you are studying (a cell, a reaction). Everything else is the surroundings. A **closed** system exchanges energy but not matter with its surroundings. Living cells are actually **open** systems (exchange both matter and energy).

JOT THIS DOWN

Free energy = energy available to do work. **ΔG** predicts feasibility.

Spontaneous if free energy decreases (ΔG < 0). Spontaneous ≠ fast. Enzymes don't change ΔG.

Cells are isothermal (constant T and P) → they use **chemical** free energy, not heat.

Photosynthetic cells ← sunlight; **heterotrophic cells** ← nutrient molecules. Both convert it to ATP.

1st law: energy is conserved (changes form, cannot be created/destroyed). **2nd law:** entropy of the universe increases.

PART 3 | State Functions: G, H and S (and ΔG = ΔH − TΔS)

THE IDEA IN PLAIN ENGLISH

A **state function** depends **only on the start and end states**, not on the route taken. Your altitude gain climbing a hill is the same whichever path you take. G, H and S are all state functions, so we can talk about their **changes (Δ)** without worrying how the reaction happens.

3.1 Gibbs free energy (G)

G = the energy capable of doing work during a process at **constant temperature and pressure**. We use **ΔG** = G(products) − G(reactants).

ΔG	Process is…	Name	What it means
Negative (< 0)	Spontaneous	**Exergonic**	Releases free energy; can do work
Positive (> 0)	Non-spontaneous	**Endergonic**	Needs free energy to be put in
Zero	At **equilibrium**	—	No net change; forward and reverse rates equal

WATCH OUT: EXAM TRAP

A living cell is **never** at equilibrium: ΔG = 0 for all its reactions would mean no net work and death. Cells stay away from equilibrium by constantly feeding in energy.

3.2 Enthalpy (H)

THE IDEA IN PLAIN ENGLISH

Enthalpy = the heat content of the system. **ΔH** is the heat absorbed or given out during a chemical process **at constant pressure**.

ΔH negative → the process **produces heat** → **exothermic** (think “exit”: heat exits). **ΔH positive** → the process **absorbs heat** → **endothermic** (heat enters).

3.3 Entropy (S)

THE IDEA IN PLAIN ENGLISH

Entropy = a quantitative measure of randomness or disorder. Ice (ordered) melting to water (more random) increases entropy. A neat stack of books falling over increases entropy.

ΔS positive → disorder of the system **increased**. **ΔS negative** → disorder **decreased**.

The slide adds that entropy relates to “heat dispersed or transferred”: heat spreading out into the surroundings makes them more disordered. (Formally ΔS = heat transferred ÷ temperature.)

3.4 The Gibbs equation

$$\Delta G = \Delta H - T\Delta S$$

- **ΔG** = change in Gibbs free energy; **ΔH** = change in enthalpy; **ΔS** = change in entropy.
- **T = absolute temperature in Kelvin: K = 273 + °C** (more exactly 273.15). Never put °C in this equation.
- Valid for biological systems because they run at **constant T and P**.
- Units: ΔG and ΔH in kJ/mol (or kcal/mol); ΔS in kJ/(mol·K), so **TΔS** comes out in kJ/mol too. If ΔS is given in J, divide by 1000 first.

THE IDEA IN PLAIN ENGLISH

The equation is a tug-of-war between two drives. Systems like to **lose heat** (ΔH negative helps spontaneity) and to **gain disorder** (ΔS positive helps spontaneity, and more so at high T because of the −TΔS term). ΔG is the net result.

ΔH	ΔS	ΔG = ΔH − TΔS	Spontaneous?
− (exothermic)	+ (more disorder)	Always negative	**Always**
+ (endothermic)	− (less disorder)	Always positive	**Never**
−	−	Negative only when T is low	At **low** temperature
+	+	Negative only when T is high	At **high** temperature

WORKED EXAMPLE

Example A. ΔH = −45 kJ/mol, ΔS = +0.12 kJ/(mol·K) at 37 °C. T = 273 + 37 = 310 K.

ΔG = −45 − (310)(0.12) = −45 − 37.2 = **−82.2 kJ/mol** → exergonic, spontaneous.

Example B (endothermic yet spontaneous). ΔH = +30 kJ/mol, ΔS = +0.10 kJ/(mol·K), T = 300 K: ΔG = 30 − (300)(0.10) = **0** → equilibrium. At any higher T, TΔS > 30 so ΔG < 0. Spontaneous when T > ΔH/ΔS = 300 K.

Example C (find ΔS). ΔG = −12 kJ/mol, ΔH = +5 kJ/mol, T = 310 K. ΔS = (ΔH − ΔG)/T = (5 + 12)/310 = **+0.055 kJ/(mol·K)**. The reaction is entropy-driven.

JOT THIS DOWN

G (Gibbs free energy) = energy available to do work at constant T, P. **ΔG < 0** spontaneous, **exergonic**; **ΔG > 0** non-spontaneous, **endergonic**; **ΔG = 0** equilibrium.

H (enthalpy) = heat content (heat transferred at constant P). **ΔH < 0 exothermic**; **ΔH > 0 endothermic**.

S (entropy) = randomness/disorder. **ΔS > 0** disorder up; **ΔS < 0** disorder down.

ΔG = ΔH − TΔS, T in Kelvin (K = 273 + °C), constant T and P.

Exothermic + entropy up = always spontaneous. Endothermic + entropy down = never.

PART 4 | Redox Potential

THE IDEA IN PLAIN ENGLISH

Oxidation = losing electrons. Reduction = gaining electrons (OIL RIG: Oxidation Is Loss, Reduction Is Gain). They always happen together (redox).

Redox potential (E) is a number that says **how strongly a compound wants to give away or grab electrons** (the slide: "the tendency that a compound has to give or receive electrons"). It is measured in volts.

- **More negative E** → compound gives up electrons easily (a strong **reducing** agent), e.g. NADH.
- **More positive E** → compound grabs electrons strongly (a strong **oxidising** agent), e.g. oxygen.
- Electrons flow **from lower (more negative) E to higher (more positive) E**. That is why energy is released when NADH passes electrons down the respiratory chain to O_2.

The equation on the slide

$$\mathbf{E = E_0 + (0.0592 / n) \cdot \log ([reducing\ agent] / [oxidising\ agent])}$$ ← as written on the slide

- $\mathbf{E_0}$ = standard redox potential (when both forms are at 1 M). **n** = number of electrons transferred. **0.0592** is 2.303RT/F at 25 °C (in volts).
- **The idea:** the actual potential E shifts away from E_0 depending on how much of each form is present (the ratio of the two forms).

WATCH OUT: EXAM TRAP

Check the textbook version of this formula. The standard Nernst equation (for a reduction half-reaction) at 25 °C is $\mathbf{E = E_0 + (0.0592/n) \log ([oxidised] / [reduced])}$, with **oxidised on top**. The slide has the ratio the other way round. In an exam, write the form your lecturer used, but know which is standard so you can spot the discrepancy.

THE IDEA IN PLAIN ENGLISH

In bioenergetics the slide says redox potential is the **ratio of NAD^+ to NADH + H^+** (the slide prints "$NADH^+$", a typo). It describes **how much NAD^+ is available for metabolism**.

Why it matters: dehydrogenases in catabolism need NAD^+ to accept electrons ($NAD^+ + 2H \rightarrow NADH + H^+$). If NAD^+ runs low (ratio falls), catabolic pathways slow down. A high NAD^+/NADH ratio = plenty of "electron acceptor" ready.

BEYOND THE SLIDE (still examinable)

Link to energy: $\Delta G^{\circ\prime} = -nF\Delta E^{\circ\prime}$ ($F = 96.485\ kJ \cdot V^{-1} \cdot mol^{-1}$).

Example: NADH → O_2. $E^{\circ\prime}(NAD^+/NADH) = -0.32$ V, $E^{\circ\prime}(O_2/H_2O) = +0.82$ V. $\Delta E^{\circ\prime} = 0.82 - (-0.32) = 1.14$ V, n = 2. $\Delta G^{\circ\prime} = -2 \times 96.485 \times 1.14 \approx$ **−220 kJ/mol**. That large release of free energy is what the respiratory chain harvests to make ATP.

JOT THIS DOWN

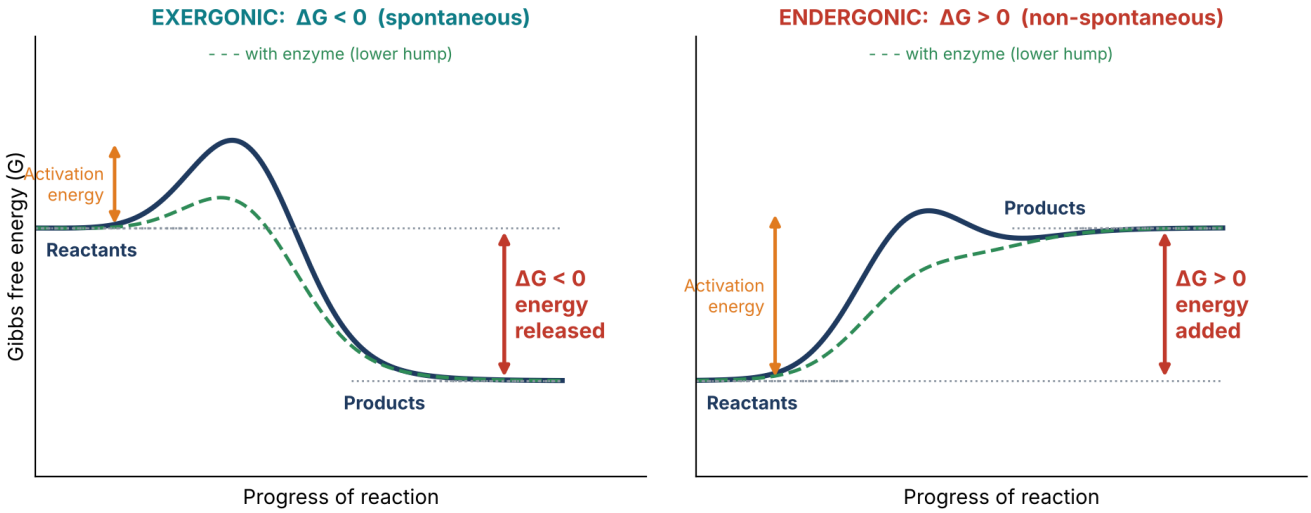
Redox potential = tendency of a compound to give or receive electrons (volts). Oxidation = loss of e^-; reduction = gain of e^-.

More negative E = better electron donor (reducing agent). More positive E = better electron acceptor.

Slide equation: $E = E_0 + (0.0592/n) \log([\text{reducing}]/[\text{oxidising}])$. (Textbook Nernst has [oxidised]/[reduced].)

In bioenergetics: ratio of **NAD^+ : NADH + H^+** = availability of NAD^+ for metabolism.

PART 5 | Exergonic and Endergonic Reactions



Energy diagrams for the two reaction types. The green dashed line shows an enzyme lowering the activation-energy hump; it does not change ΔG.

5.1 Exergonic reactions (ΔG < 0)

THE IDEA IN PLAIN ENGLISH

Exergonic = "energy going out". The reaction **releases energy** and runs **spontaneously** without needing energy input. The reactants sit **higher** on the free-energy scale than the products, so the reaction slides downhill, like a ball rolling off a table.

These reactions **can perform work** and include **most catabolic reactions** of cellular respiration (the energy released is captured as ATP).

$$\Delta G = G_{products} - G_{reactants} < 0 \text{ (ΔG is negative)}$$

- Most involve **breaking bonds** while forming reaction intermediates.
- Energy is released to the surroundings, so ΔH is **negative** (the slide's statement).
- The slide says entropy is **increased**, and that exergonic reactions include **exothermic reactions**.

5.2 Endergonic reactions (ΔG > 0)

THE IDEA IN PLAIN ENGLISH

Endergonic = "energy going in". The opposite of exergonic: **non-spontaneous** and **needs a free-energy input**. Reactants sit **lower** than the products, so the reaction is uphill, like pushing a ball up a slope.

Most anabolic reactions (photosynthesis, DNA synthesis, protein synthesis) are endergonic.

$$\Delta G = G_{products} - G_{reactants} > 0 \text{ (ΔG is positive)}$$

- Energy must be supplied from outside for the reaction to progress.
- The slide says entropy of the system is **decreased** (new, more ordered products form) and that endergonic reactions include **endothermic reactions**.

5.3 Side-by-side comparison (slide 15, tidied up)

Feature	Endergonic	Exergonic
Gibbs free energy change	**Positive** ($\Delta G > 0$)	**Negative** ($\Delta G < 0$)
Spontaneity	Non-spontaneous	Spontaneous
Energy of reactants vs products	Reactants **lower** than products	Reactants **higher** than products
Energy flow	Absorbs energy from the surroundings	Releases energy to the surroundings
Entropy (slide's statement)	Decreased	Increased
Heat (slide's statement)	Endothermic reactions are endergonic	Exothermic reactions are exergonic
Typical metabolism	Anabolic (synthesis)	Catabolic (breakdown)
What the cell does	Must be **coupled** to an exergonic reaction (e.g. ATP hydrolysis)	Can **drive** endergonic reactions; makes ATP

WATCH OUT: EXAM TRAP

The slide says exergonic reactions "do not need energy to begin" and endergonic ones "always require energy to begin". Careful: **both** types have an **activation-energy barrier** (the hump in the diagrams) that must be overcome to start. What differs is the *net* result: exergonic reactions release more than they needed to start; endergonic ones need a **net** input. Enzymes lower the hump for both.

The "exergonic = exothermic, entropy up" and "endergonic = endothermic, entropy down" links are rules of thumb. Strictly, the sign of ΔG comes from $\Delta G = \Delta H - T\Delta S$ (see Example B and C in Part 3). Write the slide's version in the exam, but know it is a simplification.

5.4 Coupled reactions: how cells pay for uphill reactions

THE IDEA IN PLAIN ENGLISH

Free-energy changes are **additive**. A reaction with a positive ΔG can be driven forward if it is **coupled** (shares a step) with a reaction that has a larger negative ΔG, so the **sum is negative**. ATP hydrolysis is the usual partner.

WORKED EXAMPLE

Glucose + Pi → glucose-6-phosphate + H_2O $\Delta G^{\circ\prime}$ = **+13.8 kJ/mol** (endergonic, will not go alone)

ATP + H_2O → ADP + Pi $\Delta G^{\circ\prime}$ = **−30.5 kJ/mol**

Coupled (hexokinase): glucose + ATP → glucose-6-phosphate + ADP $\Delta G^{\circ\prime}$ = +13.8 + (−30.5) = **−16.7 kJ/mol** (now spontaneous).

BEYOND THE SLIDE (still examinable)

ΔG vs ΔG°′: ΔG°′ is the *standard* free-energy change (1 M, pH 7, 25 °C). The **real** ΔG in a cell depends on actual concentrations: $\Delta G = \Delta G^{\circ\prime} + RT \ln([\text{products}]/[\text{reactants}])$. It is the **actual ΔG** that decides direction in the cell, and at equilibrium ΔG = 0.

Thermodynamics vs kinetics: ΔG tells you *whether* a reaction can go; activation energy and enzymes decide *how fast*.

JOT THIS DOWN

Exergonic: ΔG < 0, spontaneous, releases energy, reactants above products, **catabolic** (respiration), exothermic (slide), entropy ↑. $\Delta G = G_{prod} - G_{react} < 0$.

Endergonic: ΔG > 0, non-spontaneous, needs energy input, reactants below products, **anabolic** (photosynthesis, DNA/protein synthesis), endothermic (slide), entropy ↓.

Both have activation energy; enzymes lower it, never change ΔG.

Endergonic reactions are driven by **coupling** to ATP hydrolysis (ΔG values add). Example: +13.8 + (−30.5) = −16.7 kJ/mol.

PART 6 | High-Energy Compounds

THE IDEA IN PLAIN ENGLISH

Some compounds are like **tightly compressed springs**: when you cut them (by **hydrolysis**, splitting with water) they release a lot of free energy. These are **high-energy (energy-rich) compounds**. The cell uses them as energy carriers and to pass energy to ATP.

- **High-energy compound:** releases **at least 7 kcal/mol (about 29 kJ/mol) at pH 7.0** on hydrolysis (the slide prints “7 Cal/mol”; see Part 8).
- **Low-energy compound:** releases **less than 7 kcal/mol** (less than ATP → ADP + Pi).
- **Most high-energy compounds contain a phosphate group** (so they are called **high-energy phosphates**). **Exception: acetyl CoA** (a thioester).
- **Symbol “~”:** the squiggle in A–P~P~P marks a high-energy bond. It does not mean the bond itself is strong; it means **hydrolysing it releases a lot of energy**.

6.1 Classification of high-energy compounds

Class	Bond	Examples
Pyrophosphates	Phosphoanhydride bond (P~P)	**ATP**, pyrophosphate (PPi)
Acyl phosphates	Carboxyl–phosphate: R–C(=O)–O~P	1,3-Bisphosphoglycerate, carbamoyl phosphate, acetyl phosphate
Enol phosphates	Enol–phosphate: C=C–O~P	**Phosphoenolpyruvate (PEP)**
Thiol esters (thioesters)	Acyl–sulfur: R–C(=O)~S–CoA	Acetyl CoA, acyl CoA
Guanido phosphates (phosphagens)	N~P (phosphate on a guanidino nitrogen)	**Phosphocreatine**, phosphoarginine

JOT THIS DOWN

Memory trick for the five classes: Pyrophosphates, **A**cyl phosphates, **E**nol phosphates, **T**hiol esters, **G**uanido phosphates = "**P**lease **A**sk **E**very **T**eacher **G**enuine questions".

6.2 Why are some bonds "high-energy"?

BEYOND THE SLIDE (still examinable)

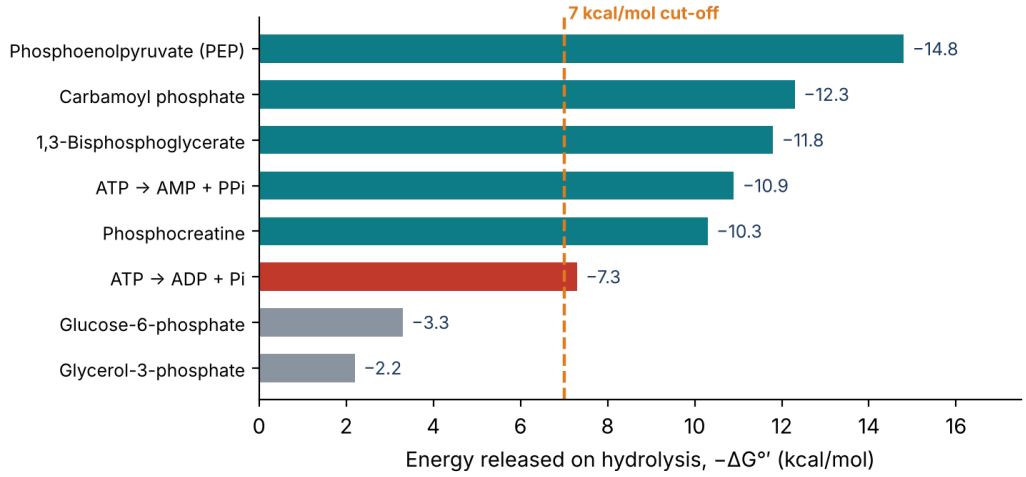
The energy does not hide in the bond. It comes from the **products being much more stable** than the reactant. Three reasons:

1. Electrostatic repulsion: the crowded negative charges on ATP's phosphates repel each other. Splitting off a phosphate relieves the strain.

2. Resonance stabilisation: free Pi has more ways to spread its charge than when it is attached, so the product is more stable.

3. Ionisation and hydration: the products ionise and are better solvated by water than the reactant.

6.3 The energy ladder and why ATP is in the middle



Approximate standard free energy of hydrolysis (textbook values; they differ slightly between books).

THE IDEA IN PLAIN ENGLISH

Compounds **above** ATP on the ladder (PEP, 1,3-bisphosphoglycerate, phosphocreatine…) have **more** energy to give, so they can **hand their phosphate to ADP and make ATP** (this is substrate-level phosphorylation). Compounds **below** ATP (glucose-6-phosphate…) **cannot** make ATP, but ATP can give them its phosphate.

ATP sits in the middle, so it works as the perfect go-between: easy to charge up from the high-energy donors, and able to charge down the low-energy acceptors. This is why ATP is the universal "currency".

WATCH OUT: EXAM TRAP

The slide says "all high-energy compounds, when hydrolysed, liberate more energy than ATP". **Not strictly:** ATP itself is a high-energy compound (7.3 kcal/mol), so the cut-off is ≥ 7. Only the compounds *above* ATP on the ladder release more.

JOT THIS DOWN

High-energy compound = releases **≥ 7 kcal/mol** (≈ 29 kJ/mol) on hydrolysis at pH 7. Low-energy = < 7.

Most are **phosphates** (high-energy phosphates). **Exception: acetyl CoA.**

5 classes: pyrophosphates (ATP, PPi); acyl phosphates (1,3-BPG, carbamoyl-P, acetyl-P); enol phosphates (PEP); thiol esters (acetyl CoA, acyl CoA); guanido phosphates (phosphocreatine, phosphoarginine).

ATP is in the **middle** of the energy ladder: accepts phosphate from higher compounds, donates to lower ones.

Memory: Please Ask Every Teacher Genuine questions.

PART 7 | Adenosine Triphosphate (ATP)

7.1 What ATP is

- **Adenosine-5′-triphosphate** is a multifunctional **nucleotide** used in cells as a **coenzyme**.
- Called the **“molecular unit of currency”** of intracellular energy transfer: it **transports chemical energy within cells** for metabolism.
- **Produced** by **photophosphorylation** and **cellular respiration**; **used** by enzymes and structural proteins in many cellular processes.
- One ATP has **three phosphate groups**. It is made by the enzyme **ATP synthase** from **inorganic phosphate (Pi) and ADP** (the slide adds “or AMP”; see Part 8).

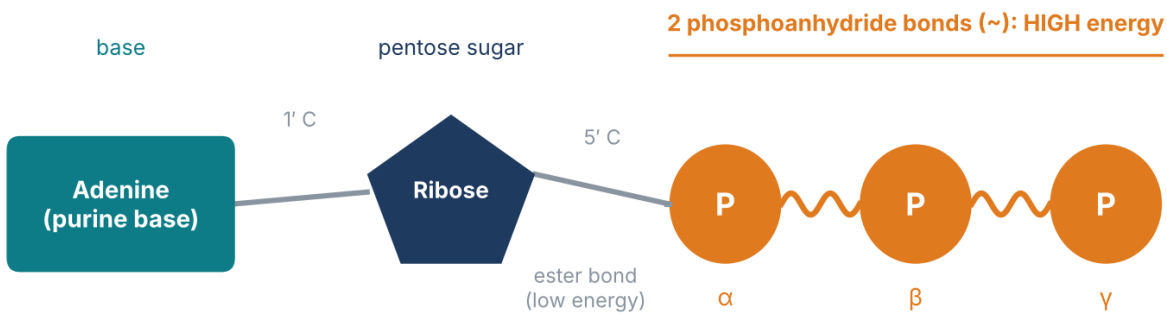
BEYOND THE SLIDE (still examinable)

ATP is not stored, it is recycled. A person has only tens of grams of ATP at any moment but turns over roughly their **own body weight in ATP every day**. Each ATP molecule is made and broken down thousands of times a day.

Phosphocreatine is the short-term backup in muscle: it quickly re-phosphorylates ADP to ATP (creatine phosphate + ADP → creatine + ATP) during sudden bursts of activity.

Adenylate kinase converts two ADP into ATP + AMP when the cell is really low on energy.

7.2 Structure of ATP



ATP − γ phosphate (+ H_2O) = ADP + Pi ATP − β,γ (as PPi) = AMP + PPi

ATP = adenine + ribose + three phosphates. The α, β, γ phosphates are counted from the sugar outward.

- A **purine base (adenine)** is attached to the **1′ carbon** of a **pentose sugar (ribose)**. Together, adenine + ribose = **adenosine** (a nucleoside).
- **Three phosphate groups** are attached to the **5′ carbon** of the ribose.
- **Adding or removing phosphate groups inter-converts ATP, ADP and AMP**: ATP (3 P) ⇄ ADP (2 P) ⇄ AMP (1 P).

WATCH OUT: EXAM TRAP

Remember the numbers: **base on 1′ carbon, phosphates on 5′ carbon.** Adenine is a **purine** (double ring). ATP has **two** high-energy phosphoanhydride bonds (between the three phosphates); the bond holding the first phosphate to the ribose is an ordinary ester.

7.3 Energy released when ATP is hydrolysed

Reaction (standard state, 1 M)	ΔG°′ (kJ/mol)	ΔG°′ (kcal/mol)
ATP + H_2O → ADP + Pi	**−30.5**	**−7.3**
ATP + H_2O → AMP + PPi	**−45.6**	**−10.9**

- **Conversion:** 1 kcal = 4.184 kJ (7.3 × 4.184 = 30.5 ✓).
- The slide says these values can be used to calculate the energy change **under physiological conditions** using the cellular **ATP/ADP ratio** (energy charge). In real cells the ATP/ADP ratio is far above equilibrium, so the actual ΔG is **more negative than the standard value (about −50 to −65 kJ/mol)**. That is why a healthy cell's ATP is such a strong energy source.

7.4 Three main functions of ATP

Function	Example
1. Transport through the cell membrane (active transport)	Moving sodium, calcium and potassium ions (e.g. the Na^+/K^+ pump)
2. Synthesis of chemical compounds	Proteins, cholesterol, DNA (anabolism)
3. Mechanical work	Muscle contraction, movement of cilia and flagella, chromosome movement

JOT THIS DOWN

ATP = adenosine-5′-triphosphate, a nucleotide/coenzyme. "Molecular unit of currency" of energy transfer.

Made by photophosphorylation and respiration (ATP synthase: ADP + Pi → ATP).

Structure: adenine (purine) on **1′ C** of ribose; **3 phosphates on 5′ C**. 2 high-energy bonds.

ATP + H_2O → ADP + Pi: ΔG°′ = −30.5 kJ/mol (−7.3 kcal/mol). ATP + H_2O → AMP + PPi: −45.6 kJ/mol (−10.9 kcal/mol).

3 functions: transport of ions across membranes; synthesis (protein, cholesterol); mechanical work (muscle contraction).

ATP/ADP ratio = **energy charge**.

PART 8 | Corrections and Traps in the Slides (pen these in)

These are the places where a slide is loose, outdated or wrong. If your exam follows the slide, write what the slide says, but make sure you know the accurate version.

Slide	What it says	Accurate version
Redox potential equation	$E = E_0 + (0.0592/n)$ log([reducing]/[oxidising])	Standard Nernst: $E = E_0 + (0.0592/n)$ log([**oxidised**]/[**reduced**]) at 25 °C.
Redox potential	"$NADH^+ + H^+$"	It is **NADH + H^+** (NADH has no + sign).
High-energy compounds	"7 Cal/mol"	**7 kcal/mol** (capital Cal = kcal). In kJ: about 29 kJ/mol.
High-energy compounds	"All liberate more energy than ATP"	Cut-off is **≥ 7 kcal/mol**, and ATP (7.3) is itself one. Only compounds above ATP release more.
Endergonic vs exergonic table	Exergonic "do not need energy to begin", endergonic "always require energy"	**Both** need activation energy. Endergonic reactions need a **net** energy input.
Exergonic/endergonic	Exergonic = exothermic + entropy up; endergonic = endothermic + entropy down	Rules of thumb. True link is $\Delta G = \Delta H - T\Delta S$ (an endothermic reaction can be spontaneous if entropy rises enough).
Entropy	"Measures the amount of heat dispersed"	Entropy measures **disorder**; ΔS = heat transferred ÷ T.
Introduction (contd)	"Goal of metabolic and catabolic processes"	Metabolism = **catabolism + anabolism**. Goals: make ATP, use ATP.
Energy charge	= ATP:ADP ratio	Simplified. Formal: ([ATP] + ½[ADP]) / ([ATP] + [ADP] + [AMP]).
ATP slide	ATP synthase makes ATP "from Pi and ADP **or AMP**"	ATP synthase uses **ADP + Pi**. AMP is first converted (e.g. adenylate kinase).
ATP functions	Transporting "**organic** substances such as sodium, calcium, potassium"	These are **inorganic ions**.
Classification table	Thiol ester drawn "–C–O~S–"; pyrophosphate drawn "–C–P–P"	Thioester is R–C(=O)~**S**–CoA; pyrophosphate is a **P~P** anhydride bond. Treat the slide's bond sketches as shorthand.
2nd law	Entropy increases "unless energy-requiring processes counteract"	Entropy of the **universe** always increases. Cells make a *local* decrease by spending energy and increasing entropy elsewhere.
Temperature	K = 273 + °C	Exactly 273.15, but 273 is accepted.

PART 9 | Test Yourself

Cover the answers (next page). Questions 1 to 17 are theory, 18 to 24 are calculations.

1. Define bioenergetics. State its goal.

2. Distinguish catabolic from anabolic reactions. Give two examples of each and say how each relates to ATP.

3. Why is ATP called the “energy currency” of the cell?

4. What is the energy charge of a cell? What does the cell do if there is more ADP than ATP? What if there is more ATP than ADP?

5. Define free energy. What does ΔG tell you, and what does it NOT tell you?

6. Why are cells described as isothermal systems? Name the energy source of (a) photosynthetic and (b) heterotrophic cells.

7. State the first and second laws of thermodynamics. How can a cell stay highly ordered without breaking the second law?

8. Define Gibbs free energy, enthalpy and entropy. State what a positive and a negative change means for each.

9. Write the equation that links ΔG, ΔH and ΔS. State the conditions and units (what is T?).

10. For which signs of ΔH and ΔS is a reaction (a) always spontaneous, (b) never spontaneous?

11. Define redox potential. What does the NAD^+/NADH ratio describe in bioenergetics?

12. Give five differences between exergonic and endergonic reactions.

13. Why are most anabolic reactions endergonic, and how does the cell drive them?

14. Define a high-energy compound. State the energy cut-off and name one example of each of the five classes.

15. Why is ATP in the “middle” of the high-energy compound ladder, and why does that matter?

16. Describe the structure of ATP (which carbons hold the base and phosphates?).

17. Write the two ATP hydrolysis reactions with their $\Delta G^{\circ\prime}$ values. State three functions of ATP.

18. A reaction has $\Delta H = -45$ kJ/mol and $\Delta S = +0.12$ kJ/(mol·K). Calculate ΔG at 37 °C. Is it exergonic or endergonic?

19. A reaction has $\Delta H = +30$ kJ/mol and $\Delta S = +0.10$ kJ/(mol·K). At what temperature does it become spontaneous?

20. Convert 37 °C to kelvin and −7.3 kcal/mol to kJ/mol.

21. Glucose + Pi → glucose-6-phosphate has $\Delta G^{\circ\prime} = +13.8$ kJ/mol. Calculate the overall $\Delta G^{\circ\prime}$ when coupled to ATP → ADP + Pi (−30.5 kJ/mol). Is the coupled reaction spontaneous?

22. A reaction has $\Delta G^{\circ\prime} = -20$ kJ/mol, but under cell conditions its actual $\Delta G = +5$ kJ/mol. Which direction does it proceed in the cell?

23. Calculate $\Delta G^{\circ\prime}$ for NADH → O_2 given $E^{\circ\prime}(NAD^+/NADH) = -0.32$ V, $E^{\circ\prime}(O_2/H_2O) = +0.82$ V, $n = 2$, $F = 96.485$ kJ/(V·mol).

24. A reaction has $\Delta G = -12$ kJ/mol and $\Delta H = +5$ kJ/mol at 310 K. Calculate ΔS. Explain how a reaction can be exergonic and endothermic.

Answers

1. **Bioenergetics** is the study of energy changes in biological reactions, the field of biochemistry concerned with the transformation and use of energy by living cells. **Goal:** to describe how organisms **acquire, transform and utilize** energy to perform biological work.

2. **Catabolic:** breakdown of molecules, releases energy, **makes ATP** (glycolysis, cellular respiration). **Anabolic:** synthesis of compounds, needs energy, **uses ATP** (protein synthesis, DNA synthesis, photosynthesis).

3. Cells transfer energy from energy-yielding reactions to energy-requiring reactions through ATP. It is made where energy is available and spent where energy is needed, as money is, and it is constantly recycled (ATP ⇄ ADP + Pi).

4. **Energy charge** = the ratio of ATP to ADP. **More ADP than ATP:** the cell must **synthesize ATP** (oxidative phosphorylation). **More ATP than ADP:** the cell **can use ATP to do work**.

5. **Free energy** = the energy actually available to do work. ΔG predicts **feasibility/spontaneity** (negative = can occur spontaneously). It does **not** tell you **how fast** the reaction goes (that is kinetics, set by activation energy and enzymes).

6. Cells work at **constant temperature and pressure**, so they cannot use heat flow to do work and rely on chemical free energy. (a) **Solar radiation**; (b) **nutrient molecules**.

7. **1st:** energy cannot be created or destroyed, only changed in form or moved (conservation of energy). **2nd:** in natural processes the entropy (disorder) of the universe increases. Cells stay ordered by **spending energy (ATP)** and releasing heat/waste, so the **total** entropy of cell + surroundings still increases.

8. **G:** energy available to do work at constant T, P (ΔG < 0 spontaneous/exergonic; > 0 non-spontaneous/endergonic; 0 equilibrium). **H:** heat content (ΔH < 0 exothermic, > 0 endothermic). **S:** disorder (ΔS > 0 disorder increased, < 0 decreased).

9. **$\Delta G = \Delta H - T\Delta S$.** Valid at constant temperature and pressure. T = absolute temperature in **kelvin** (K = 273 + °C). ΔG, ΔH in kJ/mol; ΔS in kJ/(mol·K).

10. (a) ΔH negative and ΔS positive: **always spontaneous**. (b) ΔH positive and ΔS negative: **never spontaneous**. (If both have the same sign it depends on T.)

11. **Redox potential** = a quantitative expression of a compound's tendency to give or receive electrons. In bioenergetics it is the **ratio of NAD^+ to NADH + H^+**, describing the **availability of NAD^+ for metabolism**.

12. Any five of: ΔG positive vs negative; non-spontaneous vs spontaneous; products higher vs lower in energy than reactants; absorb vs release energy; entropy decreased vs increased; endothermic vs exothermic; anabolic vs catabolic; needs a coupled energy source vs can drive other reactions.

13. Anabolic reactions build larger, more ordered molecules (entropy decreases, energy stored), so ΔG > 0. The cell **couples** them to exergonic reactions, mainly **ATP hydrolysis**, so that the overall ΔG is negative.

14. Substance releasing **at least 7 kcal/mol (≈ 29 kJ/mol) at pH 7.0** on hydrolysis. Classes: pyrophosphate (ATP), acyl phosphate (1,3-BPG), enol phosphate (PEP), thiol ester (acetyl CoA), guanido phosphate (phosphocreatine).

15. ATP's free energy of hydrolysis (7.3 kcal/mol) lies between higher-energy donors (PEP, 1,3-BPG, phosphocreatine) and lower-energy phosphate esters (glucose-6-phosphate). So it can be **recharged** by the donors and **give phosphate** to the acceptors, making it a universal energy carrier.

16. **Adenine** (purine base) is attached to the **1′ carbon** of **ribose** (pentose sugar); **three phosphate groups** are attached to the **5′ carbon**.

17. ATP + H_2O → ADP + Pi, $\Delta G^{\circ\prime}$ = **−30.5 kJ/mol (−7.3 kcal/mol)**. ATP + H_2O → AMP + PPi, $\Delta G^{\circ\prime}$ = **−45.6 kJ/mol (−10.9 kcal/mol)**. Functions: **transport** of ions across the membrane; **synthesis** of compounds (protein, cholesterol); **mechanical work** (muscle contraction).

18.	T = 310 K. ΔG = −45 − (310)(0.12) = −45 − 37.2 = **−82.2 kJ/mol**. Negative → **exergonic**.
19.	ΔG = 0 when T = ΔH/ΔS = 30/0.10 = **300 K (27 °C)**. Above 300 K, ΔG < 0, so it is **spontaneous above 300 K** (endothermic but entropy-driven).
20.	37 + 273 = **310 K** (310.15 K exactly). −7.3 × 4.184 = **−30.5 kJ/mol**.
21.	ΔG°′ = +13.8 + (−30.5) = **−16.7 kJ/mol**. Negative → **spontaneous** (coupling makes the uphill reaction go).
22.	The **actual ΔG** is what counts in the cell. ΔG = +5 means it is non-spontaneous in the forward direction, so it proceeds in the **reverse** direction under those concentrations.
23.	ΔE°′ = 0.82 − (−0.32) = 1.14 V. ΔG°′ = −nFΔE°′ = −2 × 96.485 × 1.14 = **≈ −220 kJ/mol**.
24.	ΔS = (ΔH − ΔG)/T = (5 + 12)/310 = **+0.055 kJ/(mol·K)**. The reaction absorbs heat (endothermic), but the large increase in disorder makes TΔS (17 kJ/mol) larger than ΔH (5 kJ/mol), so ΔG is negative (exergonic).

PART 10 | One-Page Revision Sheet

Topic	Key points
Bioenergetics	Study of energy changes/transformation in living cells; how organisms acquire, transform, utilize energy for work
Metabolism	Catabolic (breakdown, makes ATP) + anabolic (synthesis, uses ATP)
ATP	Energy currency. ATP ⇄ ADP + Pi. Energy charge = ATP:ADP. More ADP → oxidative phosphorylation
Free energy	Energy available to do work; ΔG < 0 spontaneous; cells isothermal; photosynthetic ← light, heterotrophic ← food
Thermodynamics	1st: energy conserved. 2nd: entropy of universe increases
G / H / S	ΔG: − exergonic, + endergonic, 0 equilibrium. ΔH: − exothermic, + endothermic. ΔS: + disorder up
Gibbs equation	**ΔG = ΔH − TΔS** (T in K = 273 + °C)
Redox potential	Tendency to give/receive e^-. Slide: $E = E_0 + (0.0592/n) \log(red/ox)$. Bioenergetics: NAD^+/NADH ratio
Exergonic	ΔG < 0, spontaneous, releases energy, catabolic, exothermic, entropy ↑, reactants > products
Endergonic	ΔG > 0, non-spontaneous, needs energy, anabolic, endothermic, entropy ↓, reactants < products
High-energy compounds	≥ 7 kcal/mol at pH 7; mostly phosphates (exception acetyl CoA). Pyrophosphates, acyl-P, enol-P, thiol esters, guanido-P
ATP structure	Adenine on 1′ C of ribose; 3 phosphates on 5′ C
ATP energy	→ ADP + Pi: −30.5 kJ/mol (−7.3 kcal). → AMP + PPi: −45.6 kJ/mol (−10.9 kcal)
ATP functions	Transport (ions), synthesis (protein, cholesterol), mechanical work (muscle)
Conversions	1 kcal = 4.184 kJ; K = 273 + °C; $\Delta G^{\circ\prime} = -nF\Delta E^{\circ\prime}$

SIX MISTAKES TO AVOID

1. Spontaneous ≠ fast; enzymes never change ΔG.
2. Use **kelvin** in ΔG = ΔH − TΔS, and match units (kJ vs J).
3. Both exergonic and endergonic reactions have an activation-energy barrier.
4. High-energy cut-off is **7 kcal/mol**, not 7 cal.
5. ATP is made from **ADP + Pi**; phosphates sit on the **5′ carbon**; the base on **1′**.
6. An endothermic reaction can still be spontaneous if entropy rises enough.